

Executive Business Report

Retail Sales Forecasting and Demand Analysis System

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Executive Summary

Efficient inventory planning is one of the most important challenges faced by retail businesses. Maintaining excess inventory increases storage costs, while insufficient inventory results in lost sales and reduced customer satisfaction. This project was developed to help businesses make informed inventory decisions by analysing historical sales data, forecasting future demand, detecting unusual sales behaviour, and grouping products based on their demand characteristics.

Historical sales records covering four years were analysed using data visualization, machine learning, anomaly detection, and clustering techniques. An interactive Streamlit dashboard was also developed so that business users can easily explore historical sales trends, forecast future demand, identify unusual sales events, and understand product demand segments without requiring any technical knowledge.

The completed system provides practical business insights that support inventory planning, purchasing decisions, demand monitoring, and stock management.

1. Business Objective

The objective of this project was to build an intelligent sales forecasting system capable of:

- Analysing historical retail sales data.
- Forecasting future sales demand.
- Detecting unusual sales behaviour.
- Segmenting products according to their demand characteristics.
- Providing an interactive dashboard for business users.

The overall goal is to assist retail managers in making better inventory and stocking decisions while reducing both overstocking and stock shortages.

2. Dataset Description

Two datasets were used during the project:

Primary Dataset

The Superstore Sales dataset contains four years of retail sales transactions, including:

- Order Date
- Product Category
- Product Sub-Category
- Region
- Sales
- Profit
- Quantity
- Customer Information

This dataset was used for sales analysis, forecasting, and product demand segmentation.

Secondary Dataset

The Video Game Sales dataset was included to demonstrate the ability to work with multiple datasets and perform multi-source business analysis.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand overall business performance before developing forecasting models.

The analysis revealed several important business insights:

- Sales increased over multiple years, indicating business growth.
- Monthly sales followed seasonal patterns with noticeable fluctuations.
- Product categories contributed differently to total revenue.
- Regional sales performance varied considerably across locations.
- Some months experienced exceptionally high or low sales, suggesting promotional periods or unusual business events.

These findings established a strong foundation for forecasting future sales.

4. Sales Forecasting

Time-series forecasting models were developed using historical monthly sales data.

Separate forecasting models were built for:

- Furniture

- Technology
- Office Supplies
- West Region
- East Region

The forecasting system predicts sales for the next three months, allowing inventory managers to prepare future stock requirements in advance.

Three-Month Forecast

Forecast Area	January 2019	February 2019	March 2019
Furniture	10,143.45	5,940.99	17,309.04
Technology	16,156.24	17,111.81	21,681.73
Office Supplies	22,217.07	22,210.15	23,817.33
West Region	14,060.43	14,976.06	26,887.54
East Region	16,933.03	19,684.63	15,576.39

These forecasts indicate that demand remains relatively stable overall while individual categories and regions experience different growth patterns. Such forecasts help businesses purchase inventory in advance and reduce operational uncertainty.

5. Forecast Model Performance

The forecasting models were evaluated using three standard performance metrics:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Mean Absolute Percentage Error (MAPE)

Forecast Model	MAE	RMSE	MAPE
Furniture	2732.74	2989.82	8.86%
Technology	10745.89	14195.68	25.14%
Office Supplies	3976.28	4048.76	14.47%
West Region	4455.01	5942.04	16.92%
East Region	12472.70	14919.33	33.97%

The results indicate that forecasting performance varies across categories and regions because customer demand patterns differ significantly. Despite these differences, the models provide sufficiently reliable short-term forecasts for inventory planning.

6. Anomaly Detection

Sales data was analysed using two anomaly detection techniques:

- Isolation Forest
- Z-Score Method

Both methods successfully identified unusual sales observations that differed significantly from normal sales behaviour.

Possible business reasons for these anomalies include:

- Festival sales
- Promotional campaigns
- Bulk customer purchases
- Inventory shortages
- Sudden changes in customer demand

Detecting these events enables businesses to investigate unexpected sales behaviour and improve future planning.

7. Product Demand Segmentation

K-Means Clustering was applied to group products according to their demand behaviour.

The clustering process considered:

- Total Sales
- Average Order Value
- Sales Volatility
- Sales Growth

Three demand segments were identified:

Cluster 0

Products with stable demand and consistent sales performance.

Recommended Strategy:

Maintain regular inventory levels and replenish stock periodically.

Cluster 1

Products with moderate sales and higher growth potential.

Recommended Strategy:

Monitor demand regularly and maintain moderate inventory levels.

Cluster 2

High-value products with highly variable demand.

Recommended Strategy:

Maintain safety stock and monitor inventory carefully to avoid shortages.

This segmentation allows businesses to apply different inventory strategies to different product groups instead of using a single stocking policy.

8. Interactive Dashboard

A complete Streamlit dashboard was developed to make the forecasting system accessible to business users.

The dashboard contains four sections:

Sales Overview Dashboard

- Total sales by year
- Monthly sales trend
- Interactive filters for region and category

Forecast Explorer

- Forecast by category or region
- Adjustable forecast horizon
- Forecast visualization
- Model evaluation metrics

Anomaly Report

- Isolation Forest anomaly visualization
- Z-Score anomaly visualization
- Business interpretation of detected anomalies

Product Demand Segments

- Cluster visualization
- Product cluster assignments
- Cluster summary
- Inventory recommendations

The dashboard allows managers to explore business insights without requiring programming knowledge.

9. Business Recommendations

Based on the analysis, the following recommendations are suggested:

1. Use monthly sales forecasts to improve inventory planning and reduce both overstocking and stock shortages.
2. Continuously monitor unusual sales behaviour to identify promotional effects, operational issues, or unexpected market changes.
3. Apply different inventory policies for different product demand segments rather than using a single stocking strategy for all products.

These recommendations can improve inventory efficiency, reduce operational costs, and increase customer satisfaction.

10. Risk and Limitation

The forecasting models rely on historical sales data and assume that future market conditions will follow similar patterns.

Unexpected factors such as:

- Economic changes
- Supply chain disruptions
- Government policies
- Competitor actions
- Major promotional campaigns

may reduce forecasting accuracy.

Therefore, forecast models should be retrained periodically using updated sales data to maintain reliable business predictions.

Conclusion

This project successfully demonstrates how machine learning and business analytics can support retail inventory planning. Historical sales analysis, demand forecasting, anomaly detection, product segmentation, and an interactive dashboard were integrated into a single decision-support system.

The developed solution enables retail managers to understand historical performance, anticipate future demand, identify unusual sales events, and optimise inventory strategies. With regular updates and retraining, the system can continue to support informed business decisions and improve operational efficiency.

